

Chapter 14

Functions

Only One Option Correct Type Questions

1. Let (x_0, y_0) be the solution of the following equations
$$(2x)^{\ln 2} = (3y)^{\ln 3}$$
$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is [IIT-JEE-2011 (Paper-1)]
- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$
(C) $\frac{1}{2}$ (D) 6
2. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is [IIT-JEE-2011 (Paper-2)]
- (A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ (B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
(C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ (D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
3. The function $f : [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is [IIT-JEE-2012 (Paper-1)]
- (A) One-one and onto (B) Onto but not one-one
(C) One-one but not onto (D) Neither one-one nor onto
4. If the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE? [JEE (Adv)-2020 (Paper-1)]
- (A) f is one-one, but NOT onto (B) f is onto, but NOT one-one
(C) f is BOTH one-one and onto (D) f is NEITHER one-one NOR onto

One or More Option(s) Correct Type Questions

5. Let $f : (0, 1) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{b-x}{1-bx}$ where b is a constant such that $0 < b < 1$. Then [IIT-JEE-2011 (Paper-2)]
- (A) f is not invertible on $(0, 1)$ (B) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$
(C) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0, 1)$

6. If $3^x = 4^{x-1}$, then $x =$

[JEE (Adv)-2013 (Paper-2)]

(A) $\frac{2\log_3 2}{2\log_3 2 - 1}$

(B) $\frac{2}{2 - \log_2 3}$

(C) $\frac{1}{1 - \log_4 3}$

(D) $\frac{2\log_2 3}{2\log_2 3 - 1}$

7. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then

[JEE (Adv)-2014 (Paper-1)]

(A) $f(x)$ is an odd function

(B) $f(x)$ is a one-one function

(C) $f(x)$ is an onto function

(D) $f(x)$ is an even function

8. Let $f(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$ for all $x \in \mathbb{R}$ and $g(x) = \frac{\pi}{2}\sin x$ for all $x \in \mathbb{R}$. Let $(f \circ g)(x)$ denote $f(g(x))$ and $(g \circ f)(x)$ denote $g(f(x))$. Then which of the following is (are) true?

[JEE (Adv)-2015 (Paper-1)]

(A) Range of f is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(B) Range of $f \circ g$ is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(D) There is an $x \in \mathbb{R}$ such that $(g \circ f)(x) = 1$

9. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then

[JEE (Adv)-2016 (Paper-1)]

(A) $g'(2) = \frac{1}{15}$

(B) $h'(1) = 666$

(C) $h(0) = 16$

(D) $h(g(3)) = 36$

Matrix-Match / Matrix Based Type Questions

10. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by [JEE (Adv)-2014 (Paper-2)]

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

List-I

P. f_4 is

Q. f_3 is

R. $f_2 \circ f_1$ is

S. f_2 is

List-II

1. Onto but not one-one

2. Neither continuous nor one-one

3. Differentiable but not one-one

4. Continuous and one-one

P	Q	R	S
(A) 3	1	4	2
(B) 1	3	4	2
(C) 3	1	2	4
(D) 1	3	2	4

11. Let $E_1 = \left\{ x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0 \right\}$ and $E_2 = \left\{ x \in E_1 : \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right) \text{ is a real number} \right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes value in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.)

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e \left(\frac{x}{x-1} \right)$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right)$

[JEE (Adv)-2018 (Paper-2)]

List-I

- P. The range of f is
 Q. The range of g contains
 R. The domain of f contains
 S. The domain of g is

List-II

- $\left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$
- $(0, 1)$
- $\left[-\frac{1}{2}, \frac{1}{2} \right]$
- $(-\infty, 0) \cup (0, \infty)$
- $\left(-\infty, \frac{e}{e-1} \right]$
- $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1} \right]$

The correct option is

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$ (B) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$
 (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6$ (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$

Integer / Numerical Value Type Questions

12. The value of $6 + \log_3 \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots \right)$ is

[IIT-JEE-2012 (Paper-1)]

13. The value of $\left((\log_2 9)^2 \right)^{\frac{1}{\log_2(\log_2 9)}} \times (\sqrt{7})^{\frac{1}{\log_4 7}}$ is _____.

[JEE (Adv)-2018 (Paper-1)]

14. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X , then the value of $\frac{1}{5!}(\beta - \alpha)$ is _____.

[JEE (Adv)-2018 (Paper-2)]

15. Let the function $f : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{4^x}{4^x + 2}.$$

Then the value of

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \dots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$$

is _____

[JEE (Adv)-2020 (Paper-2)]

